

IDS 702 HW 4

KEY

Instructions: Use this template to complete your assignment. When you click “Render,” you should get a PDF document that contains both your answers and code. You must show your work/justify your answers to receive credit. Submit your rendered PDF file on Gradescope. **Remember to render frequently**, as this will help you to catch errors in your code before the last minute. You must show your work for all problems, and you must provide a written answer for all problems; it is insufficient to just show the code.

Add your name in the Author section in the header

Load data

```
library(tidyverse)
library(openintro)
library(caret)
library(pROC)

data("resume")
```

1

a.

```
resume <- resume |>
  mutate(
    received_callback_fac = factor(received_callback,
                                   levels=c(0,1),
                                   labels=c("No","Yes")),
    race_fac = factor(race),
    gender_fac = factor(gender),
    college_degree_fac = factor(college_degree,
```

```

                                levels=c(0,1),
                                labels=c("No","Yes")),
  resume_quality_fac = factor(resume_quality)
)

resume |>
  count(received_callback, received_callback_fac)

```

```

# A tibble: 2 x 3
  received_callback received_callback_fac     n
      <dbl> <fct>                <int>
1           0 No                 4478
2           1 Yes                 392

```

```

resume |>
  count(race, race_fac)

```

```

# A tibble: 2 x 3
  race race_fac     n
  <chr> <fct>   <int>
1 black black   2435
2 white white   2435

```

```

resume |>
  count(gender, gender_fac)

```

```

# A tibble: 2 x 3
  gender gender_fac     n
  <chr> <fct>   <int>
1 f     f       3746
2 m     m       1124

```

```

resume |>
  count(college_degree, college_degree_fac)

```

```

# A tibble: 2 x 3
  college_degree college_degree_fac     n
      <dbl> <fct>                <int>
1           0 No                 1366
2           1 Yes                 3504

```

```
resume |>
  count(resume_quality, resume_quality_fac)
```

```
# A tibble: 2 x 3
  resume_quality resume_quality_fac     n
  <chr>          <fct>          <int>
1 high          high          2446
2 low           low           2424
```

b.

```
resume |>
  count(race_fac) |>
  mutate(prop=n/sum(n))
```

```
# A tibble: 2 x 3
  race_fac     n prop
  <fct>   <int> <dbl>
1 black   2435  0.5
2 white   2435  0.5
```

```
resume |>
  group_by(received_callback_fac) |>
  count(race_fac) |>
  mutate(prop=n/sum(n))
```

```
# A tibble: 4 x 4
# Groups:   received_callback_fac [2]
  received_callback_fac race_fac     n prop
  <fct>                <fct>   <int> <dbl>
1 No                    black   2278 0.509
2 No                    white   2200 0.491
3 Yes                   black    157 0.401
4 Yes                   white    235 0.599
```

```
resume |>
  count(gender_fac) |>
  mutate(prop=n/sum(n))
```

```
# A tibble: 2 x 3
  gender_fac      n prop
  <fct>      <int> <dbl>
1 f          3746 0.769
2 m          1124 0.231
```

```
resume |>
  group_by(received_callback_fac) |>
  count(gender_fac) |>
  mutate(prop=n/sum(n))
```

```
# A tibble: 4 x 4
# Groups:   received_callback_fac [2]
  received_callback_fac gender_fac      n prop
  <fct>                <fct>    <int> <dbl>
1 No                    f        3437 0.768
2 No                    m        1041 0.232
3 Yes                   f         309 0.788
4 Yes                   m          83 0.212
```

```
resume |>
  count(college_degree_fac) |>
  mutate(prop=n/sum(n))
```

```
# A tibble: 2 x 3
  college_degree_fac      n prop
  <fct>      <int> <dbl>
1 No          1366 0.280
2 Yes          3504 0.720
```

```
resume |>
  group_by(received_callback_fac) |>
  count(college_degree_fac) |>
  mutate(prop=n/sum(n))
```

```
# A tibble: 4 x 4
# Groups:   received_callback_fac [2]
  received_callback_fac college_degree_fac      n prop
  <fct>                <fct>    <int> <dbl>
1 No                    No        1251 0.279
```

2	No	Yes	3227	0.721
3	Yes	No	115	0.293
4	Yes	Yes	277	0.707

```
resume |>
  count(resume_quality_fac) |>
  mutate(prop=n/sum(n))
```

```
# A tibble: 2 x 3
  resume_quality_fac      n prop
  <fct>             <int> <dbl>
1 high             2446 0.502
2 low              2424 0.498
```

```
resume |>
  group_by(received_callback_fac) |>
  count(resume_quality_fac) |>
  mutate(prop=n/sum(n))
```

```
# A tibble: 4 x 4
# Groups:   received_callback_fac [2]
  received_callback_fac resume_quality_fac      n prop
  <fct>                 <fct>          <int> <dbl>
1 No                    high            2232 0.498
2 No                    low             2246 0.502
3 Yes                   high             214 0.546
4 Yes                   low             178 0.454
```

```
resume |>
  summarise(mean(years_experience), sd(years_experience))
```

```
# A tibble: 1 x 2
  `mean(years_experience)` `sd(years_experience)`
  <dbl>                  <dbl>
1      7.84              5.04
```

```
resume |>
  group_by(received_callback_fac) |>
  summarise(mean(years_experience), sd(years_experience))
```

```
# A tibble: 2 x 3
  received_callback_fac `mean(years_experience)` `sd(years_experience)`
  <fct>                  <dbl>                  <dbl>
1 No                      7.75                      4.99
2 Yes                     8.89                      5.54
```

Variable	Overall 4870	Received Callback 392 (8)	Did not receive callback 4478 (92)
Race			
Black - N (%)	2435 (50)	157 (40)	2278 (51)
White - N (%)	2435 (50)	235 (60)	2200 (49)
Gender			
Female - N (%)	3746 (77)	309 (79)	3437 (77)
Male - N (%)	1124 (23)	83 (21)	1041 (23)
Has college degree - N (%)	3504 (72)	277 (71)	3227 (72)
Resume Quality			
Low - N (%)	2424 (50)	178 (45)	2246 (50)
High - N (%)	2446 (50)	214 (55)	2232 (50)
Years of experience - mean (SD)	7.8 (5)	8.9 (5.5)	7.8 (5)

Takeaways: Most variables do not seem to have a big difference when it comes to receiving callbacks. The exception to this is race. While White applicants make up 50% of the experimental sample, they receive 60% of the callbacks. Callback rates align with the overall sample for gender. Interestingly, resume quality seems to have little effect, with low quality resumes receiving 45% of the callbacks.

2

a.

```
resume_mod1 <- glm(received_callback_fac ~
  race_fac + gender_fac + college_degree_fac +
  resume_quality_fac + years_experience,
  data=resume,
  family="binomial")

summary(resume_mod1)
```

Call:

```
glm(formula = received_callback_fac ~ race_fac + gender_fac +  
     college_degree_fac + resume_quality_fac + years_experience,  
     family = "binomial", data = resume)
```

Deviance Residuals:

Min	1Q	Median	3Q	Max
-0.7114	-0.4368	-0.3955	-0.3403	2.5010

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-2.870703	0.155563	-18.454	< 2e-16 ***
race_facwhite	0.439892	0.107573	4.089	4.33e-05 ***
gender_facm	-0.106411	0.131915	-0.807	0.420
college_degree_facYes	-0.027668	0.119050	-0.232	0.816
resume_quality_faclow	-0.153108	0.106616	-1.436	0.151
years_experience	0.037576	0.009286	4.047	5.20e-05 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 2726.9 on 4869 degrees of freedom
Residual deviance: 2690.3 on 4864 degrees of freedom
AIC: 2702.3

Number of Fisher Scoring iterations: 5

```
exp(coefficients(resume_mod1))
```

	(Intercept)	race_facwhite	gender_facm
	0.05665908	1.55253916	0.89905501
college_degree_facYes			
resume_quality_faclow	0.97271142	0.85803734	1.03829139
years_experience			

b.

```
exp(confint(resume_mod1))
```

	2.5 %	97.5 %
--	-------	--------

(Intercept)	0.04160769	0.07657934
race_facwhite	1.25878536	1.91963993
gender_facm	0.69064530	1.15904435
college_degree_facYes	0.77226785	1.23202030
resume_quality_faclow	0.69577756	1.05706389
years_experience	1.01922849	1.05704813

c.

Gender, having a college degree, and resume quality are not statistically significantly associated with receiving a resume callback ($p=0.4$, 0.8 , and 0.2 , respectively). The odds of receiving a callback are 1.6 times higher for White applicants compared to Black applicants, and this is statistically significant ($p<0.001$, 95% CI: [1.3, 1.9]). Years of experience is also significantly associated with receiving a callback; per additional year of experience, the odds of receiving a callback increase by 4% ($p<0.001$, 95% CI: [1.01, 1.06]).

3

a.

```
resume_mod2 <- glm(received_callback_fac ~
  race_fac*gender_fac + college_degree_fac +
  resume_quality_fac + years_experience,
  data=resume,
  family="binomial")

summary(resume_mod2)
```

Call:

```
glm(formula = received_callback_fac ~ race_fac * gender_fac +
  college_degree_fac + resume_quality_fac + years_experience,
  family = "binomial", data = resume)
```

Deviance Residuals:

Min	1Q	Median	3Q	Max
-0.7115	-0.4369	-0.3955	-0.3402	2.5014

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-2.870419	0.159501	-17.996	< 2e-16 ***


```

race_facwhite          0.439443    0.121104    3.629 0.000285 ***
gender_facm            -0.107688    0.206252   -0.522 0.601587
college_degree_facYes  -0.027679    0.119059   -0.232 0.816162
resume_quality_faclow  -0.153111    0.106617   -1.436 0.150979
years_experience        0.037575    0.009287    4.046 5.21e-05 ***
race_facwhite:gender_facm 0.002126    0.263788    0.008 0.993571
---

```

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

```

Null deviance: 2726.9 on 4869 degrees of freedom
Residual deviance: 2690.3 on 4863 degrees of freedom
AIC: 2704.3

```

Number of Fisher Scoring iterations: 5

```
exp(coefficients(resume_mod2))
```

```

              (Intercept)              race_facwhite              gender_facm
              0.05667517              1.55184335              0.89790738
college_degree_facYes  resume_quality_faclow  years_experience
              0.97270019              0.85803441              1.03828999
race_facwhite:gender_facm
              1.00212781

```

```
exp(confint(resume_mod2))
```

```

              2.5 %      97.5 %
(Intercept)  0.0412839 0.07716664
race_facwhite 1.2255380 1.97099631
gender_facm   0.5904001 1.32864260
college_degree_facYes 0.7722465 1.23202719
resume_quality_faclow 0.6957741 1.05706202
years_experience 1.0192243 1.05704996
race_facwhite:gender_facm 0.5999197 1.69138564

```

The reference group is Black females. Let $x_1 = 1$ for race=White, 0=Black; $x_2 = 1$ for gender=male, 0=female.

- For White males, $x_1 = 1$ and $x_2 = 1$, so the log odds of receiving a callback = $-2.87 + 0.44 - 0.11 + 0.002 = -2.5$, so the odds of receiving a callback are $\exp(-2.5) = 0.08$.

- For White females, $x_1 = 1$ and $x_2 = 0$, so the log odds of receiving a callback = $-2.87 + 0.44$, so the odds of receiving a callback are $\exp(-2.4) = 0.09$.
- For Black males, $x_1 = 0$ and $x_2 = 1$, so the log odds of receiving a callback = $-2.87 - 0.11$, so the odds of receiving a callback are $\exp(-2.98) = 0.05$
- For Black females, $x_1 = 0$ and $x_2 = 0$, so the log odds of receiving a callback = -2.87 (the intercept!), so the odds of receiving a callback are $\exp(-2.9) = 0.06$

So, White females have the highest odds of receiving a callback.

b.

```
resumemod_preds <- ifelse(predict(resume_mod2, type="response")>0.5,1,0)
resumemod_preds_fac <- factor(resumemod_preds,
                             levels=c(0,1),
                             labels=c("No", "Yes"))
confusionMatrix(table(resumemod_preds_fac, resume$received_callback_fac),
                 positive="Yes",
                 mode="everything")
```

Confusion Matrix and Statistics

resumemod_preds_fac	No	Yes
No	4478	392
Yes	0	0

Accuracy : 0.9195
95% CI : (0.9115, 0.927)

No Information Rate : 0.9195
P-Value [Acc > NIR] : 0.5134

Kappa : 0

Mcnemar's Test P-Value : <2e-16

Sensitivity : 0.00000
Specificity : 1.00000
Pos Pred Value : NaN
Neg Pred Value : 0.91951
Precision : NA
Recall : 0.00000
F1 : NA

```

Prevalence : 0.08049
Detection Rate : 0.00000
Detection Prevalence : 0.00000
Balanced Accuracy : 0.50000

```

```
'Positive' Class : Yes
```

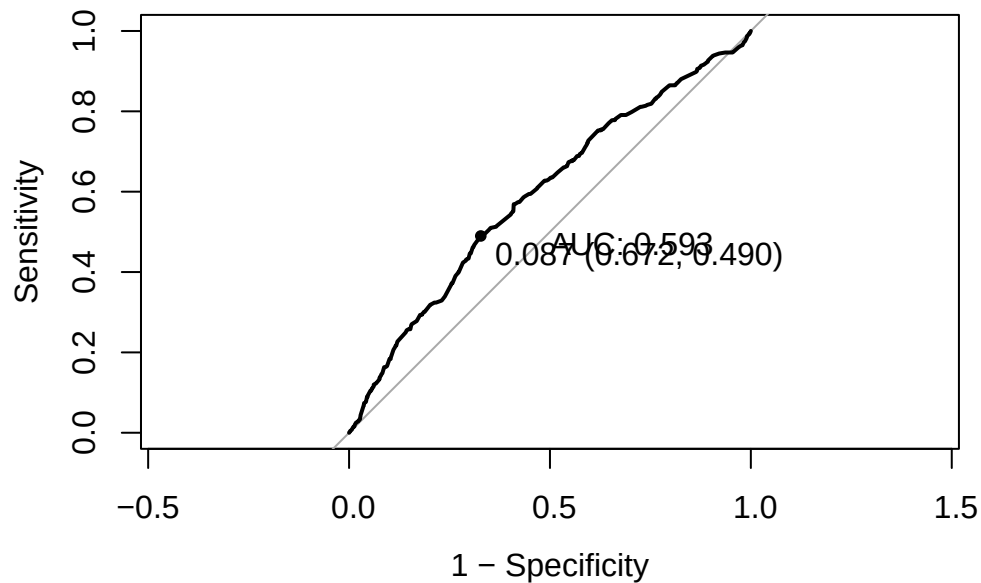
Using the 0.5 threshold, no observations are predicted to get a callback. Although the accuracy is high (0.92), the kappa is 0 due to the imbalance in the data (only 8% receive a callback).

c.

```

roc(resume$received_callback_fac, predict(resume_mod2, type="response"),
  print.thres="best",
  print.auc=T,
  legacy.axes=T,
  plot=T)

```



Call:

```
roc.default(response = resume$received_callback_fac, predictor = predict(resume_mod2, type="response"),
```

Data: predict(resume_mod2, type = "response") in 4478 controls (resume\$received_callback_fac
Area under the curve: 0.5931

```
resumemod_preds_bestthres <- ifelse(predict(resume_mod2, type="response")>0.087,1,0)
resumemod_preds_fac_bestthres <- factor(resumemod_preds_bestthres,
                                         levels=c(0,1),
                                         labels=c("No", "Yes"))
confusionMatrix(table(resumemod_preds_fac_bestthres,
                      resume$received_callback_fac),
                 positive="Yes",
                 mode="everything")
```

Confusion Matrix and Statistics

resumemod_preds_fac_bestthres	No	Yes
No	3010	200
Yes	1468	192

Accuracy : 0.6575
95% CI : (0.644, 0.6708)
No Information Rate : 0.9195
P-Value [Acc > NIR] : 1

Kappa : 0.0654

McNemar's Test P-Value : <2e-16

Sensitivity : 0.48980
Specificity : 0.67218
Pos Pred Value : 0.11566
Neg Pred Value : 0.93769
Precision : 0.11566
Recall : 0.48980
F1 : 0.18713
Prevalence : 0.08049
Detection Rate : 0.03943
Detection Prevalence : 0.34086
Balanced Accuracy : 0.58099

'Positive' Class : Yes

We now have observations that were predicted to receive a callback. The accuracy decreased, but the kappa increased. The sensitivity and specificity are fairly low at 0.49 and 0.67, respectively. Given that the AUC is 0.58, this may not be a great model for prediction, but we can still trust our inferences!

4

Based on both models (with and without interaction), race and years of experience are statistically significantly associated with receiving a callback. Gender is not significant, and the interaction between race and gender is not significant. Looking at odds of receiving a callback for each race/gender combination, we see that White females and males have odds of .09 and .08, respectively, and Black females and males have odds of 0.06 and 0.05, respectively. The strength of the model in terms of predictive accuracy is not very good; this can be difficult with a fairly rare outcome like we have in this case. Looking at the data available, it could be interesting to look at different kinds of jobs and whether or not race and gender interact with job types and receiving a callback. For example, it could be interesting to see if men are more likely to receive a callback for titles that imply leadership, like managerial roles.